

Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	5.75
NN	5.875
Sweep	5.75
SaveSeq	4.625
SaveParall	2.625
MJ	1
CMT	2.375

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 6 degrees of freedom): 40.392857.
P-value computed by Friedman Test: 0.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
6	NN	4.513373	0.000006	0.008333	0.008512	0.041946
5	Random	4.397645	0.000011	0.01	0.016952	0.041946
4	Sweep	4.397645	0.000011	0.0125	0.025321	0.041946
3	SaveSeq	3.356098	0.000791	0.016667	0.033617	0.041946
2	SaveParall	1.504458	0.132464	0.025	0.041844	0.041946
1	CMT	1.273003	0.203017	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.025 .
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.041844 .
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.041946 .

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	NN	0.000006	0.000038
2	Random	0.000011	0.000055
3	Sweep	0.000011	0.000055
4	SaveSeq	0.000791	0.002372
5	SaveParall	0.132464	0.264927
6	CMT	0.203017	0.264927

Table 3: Adjusted p -values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	NN	0.000006	0.000038	0.000008
2	Random	0.000011	0.000038	0.000014
3	Sweep	0.000011	0.000038	0.000014
4	SaveSeq	0.000791	0.001186	0.000991
5	SaveParall	0.132464	0.156772	0.142519
6	CMT	0.203017	0.203017	0.203017

Table 4: Adjusted p -values (FRIEDMAN) (II)